ARYA SABALE

COMPUTER SCIENCE AND ENGINEERING (AI & ML)
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LinkedIn | GitHub | Portfolio Pune, Maharashtra, India



SUMMARY

B.Tech Computer Science and Engineering (AI & ML) student with **9.64 CGPA**, seeking an AI Engineer Intern role. Hands-on experience developing AI-powered applications using **Python, Machine Learning, Deep Learning, NLP, Generative AI, and Large Language Models (LLMs)**. Experienced in building **RAG pipelines, AI agents, automated workflows, and API-integrated applications**. IEEE-published researcher with practical experience in retrieval evaluation, benchmarking, debugging, and backend development. Strong foundation in problem-solving, model evaluation, and software development.

EDUCATION

Bract's Vishwakarma Institute of Information Technology, Pune June 2024 – March 2027
B.Tech Computer Science and Engineering (AI & ML) | CGPA: **9.64/10.0**

Karmaveer Bhaurao Patil Polytechnic (MSBTE), Satara June 2022 – March 2024
Diploma in Computer Engineering | 92.40% | **Institute Rank 1**

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript, SQL, Object-Oriented Programming (OOP), Data Structures and Algorithms
AI/ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), Prompt Engineering, AI Agents, Langchain
ML Frameworks & Libraries: TensorFlow, Scikit-learn, Hugging Face, FastAPI
LLMs & Information Retrieval: Retrieval-Augmented Generation (RAG), Embeddings, Document Chunking, Semantic Search, FAISS, BM25, MPNet, Dense/Sparse/Hybrid Retrieval
APIs, Databases & Cloud: REST APIs, Google Docs API, Gmail API, MongoDB, MySQL, Redis, AWS (S3, EC2)
Development Practices: Git, GitHub, Debugging, Testing, Performance Evaluation, API Integration, Workflow Automation

WORK EXPERIENCE

Android Developer Intern | *Navyug Infotech Pvt. Ltd.*

1. Developed three Android applications using Java and XML, implementing application logic and user interfaces within defined project timelines.
2. Performed debugging, testing, and performance optimization to improve application stability and loading performance.

PROJECTS

RAG-Based Legal Assistant for Indian Penal Code – IEEE Published GitHub
Python, MPNet, BM25, FAISS, LLaMA-3, Mistral-7B, FastAPI, Hugging Face

1. Designed a hybrid **Retrieval-Augmented Generation (RAG) pipeline** using document chunking, MPNet embeddings, BM25 sparse retrieval, FAISS vector search, and LLM-based context generation for legal information retrieval.
2. Evaluated 36 configurations through 10,800 evaluation calls across 300 queries, achieving **73.7% Top-5 retrieval accuracy**. Documented benchmark results and retrieval performance in IEEE-published research.

Autonomous AI Business Operations Platform (NxtBiz) GitHub
AI Agents, React, Node.js, Express, MongoDB, BullMQ, Redis, Socket.IO, JWT

1. Engineered an **eight-agent AI orchestration pipeline** for intent classification, task planning, CRM, invoicing, customer support, email, and workflow automation. Classified customer emails by sentiment, intent, and urgency to trigger automated downstream actions.
2. Developed REST APIs and backend workflows across 15+ modules, implementing role-based access control, JWT authentication, and execution traceability.

Resume Checker Bot – LLM-Integrated AI Agent Live Demo
GPT-4-mini, Prompt Engineering, Make.com, Google Docs API, Gmail API

1. Developed and deployed an automated **Generative AI resume analysis agent** using GPT-4-mini to analyze resumes, generate structured feedback, and provide ratings, achieving **95% user satisfaction**.
2. Integrated Google Docs and Gmail APIs to automate document processing and user responses, reducing manual intervention in the workflow.

ACHIEVEMENTS & LEADERSHIP

1. **IEEE and Scientific Data Publication :** RP-OCNN: A Hybrid Squeeze-Excitation Residual Network for Fine-Grained Rose Petal Condition Classification. and RoseVisuals: A Multi-Class Dutch Rose Petal Images Dataset for Automated Health and Pigmentation Classification via Deep Learning.
2. **State Champion:** First place in a state-level paper presentation competition. **Academic Excellence:** Institute Rank 1 in diploma (92.40%); second rank in 10th and MSBTE examinations.
3. **Leadership:** Head of Student Body at Nirmala Convent School; organized 10+ technical and non-technical events at KBP Polytechnic.